# Practice workbook · The TCP/IP model: four layers and what each does

Naming the four layers (application, transport, network, data link) and stating what each does to packets on sending and receiving.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-04

# 1. Question 3 · part b · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b only.

**3** Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

**(2)**

| Network type | Tick (✓) | Usage model | Tick (✓) |
|---|---|---|---|
| Local Area Network (LAN) | | Client-Server | |
| Wide Area Network (WAN) | | Peer-to-peer | |
| Personal Area Network (PAN) | | | |

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

**(4)**

| | Layer name | Function |
|---|---|---|
| 1 | | |
| 2 | | |

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# 1. Question 3 · part b · 4 marks

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Worked-example retry · Attempt part b only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

**(Total for Question 3 = 13 marks)**

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## 1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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## 2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

**1** Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

**(2)**

1 .....

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2 .....

.....

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## 2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

**(2)**

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

**(2)**

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## 2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

(e) Network layouts can be based upon a number of different network topologies.

**Figure 1** shows a network topology.

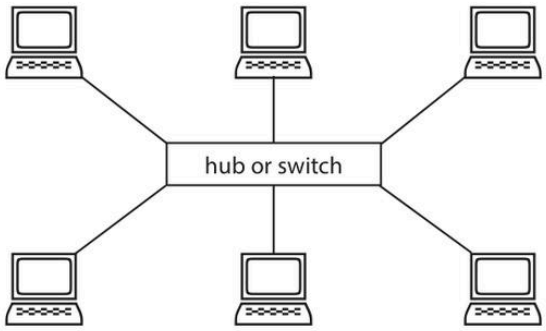



**Figure 1**

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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## 2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

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(f) The network uses TCP/IP.

**Figure 2** shows the TCP/IP protocol stack.

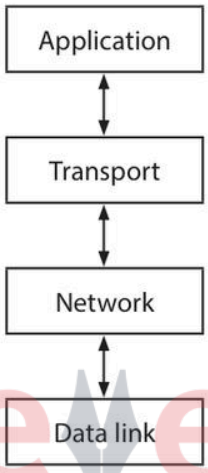



**Figure 2**

(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

**(2)**

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 1 = 17 marks)**

## 2. Question 1 · part f.ii · 2 marks · Working and answer

Working and answer

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# 3. Question 2 · part c.i · 1 mark

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Further practice · Attempt part c.i only.

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**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

.....

.....

(b) State the purpose of network protocols.

**(1)**

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

**(1)**

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

## 3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

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**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

.....

.....

(b) State the purpose of network protocols.

**(1)**

.....

.....

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(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

**(1)**

.....

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(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

## 4. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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## 5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

**1** There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

**(2)**

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

**(1)**

.....

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## 5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

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(c) **Figure 1** shows a network topology.

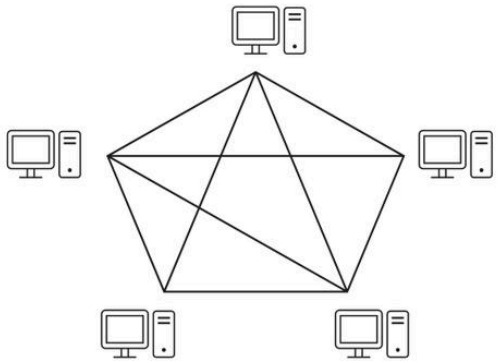


**Figure 1**

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

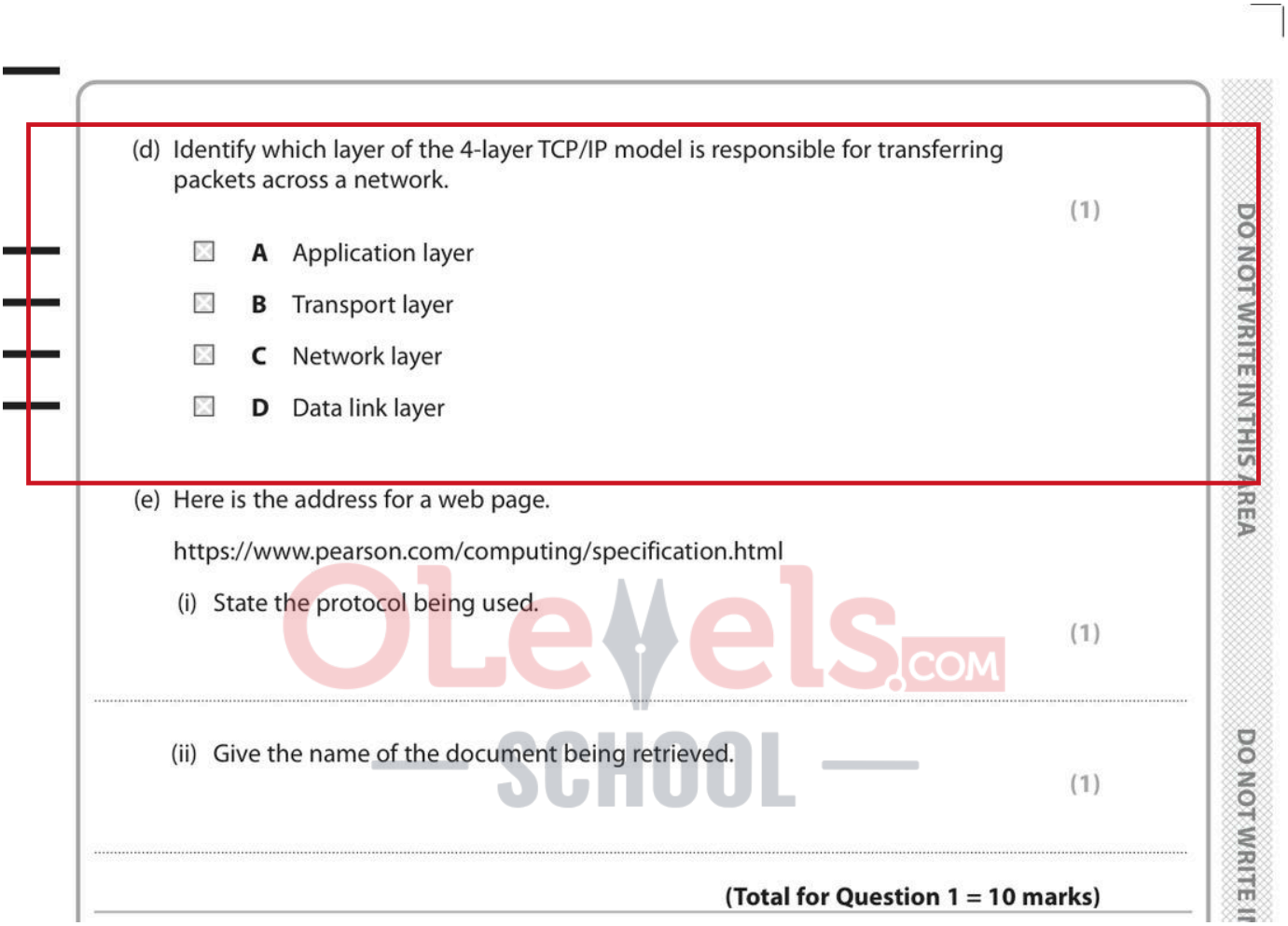
(2)

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## 5. Question 1 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.



(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

**(Total for Question 1 = 10 marks)**

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Working and answer

## 6. Question 1 · part b · 2 marks

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Further practice · Attempt part b only.

**1** Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

**(4)**

| Statement | DNS | PAN | WAN | WWW |
|---|---|---|---|---|
| A network of networks **not** limited to a single location | ☒ | ☒ | ☒ | ☒ |
| A network that connects devices within the range of a single user | ☒ | ☒ | ☒ | ☒ |
| A set of hypertext documents accessed through the HTTP protocol | ☒ | ☒ | ☒ | ☒ |
| A system that maps domain names to IP addresses | ☒ | ☒ | ☒ | ☒ |

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

**(2)**

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

**(1)**

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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## 6. Question 1 · part b · 2 marks · Working and answer

Working and answer

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